

Investigating the Impact of Distributionally Robust Optimization on Learning with Noisy Labels

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ABSTRACT Traditional deep learning models trained via Empirical Risk Minimization (ERM) often fail to generalize under real-world imperfections such as spurious correlations and noisy labels. While recent literature has addressed these challenges in isolation, developing Distributionally Robust Optimization (DRO) for group shifts and semi-supervised pipelines for noisy labels, the intersection of these problems remains underexplored. In this work, we investigate whether robust feature representations learned through Group DRO can transfer across domains and improve learning under label noise. We adopt a sequential robust transfer framework in which a ResNet-50 model is first trained on the Waterbirds dataset using Group DRO to suppress reliance on spurious background correlations. The resulting encoder is then transferred to a noisy-label setting on CIFAR-10N and fine-tuned using a DivideMix-style semi-supervised pipeline. We address the architectural mismatch between high-resolution source domains and low-resolution target domains using a selective layer transfer that preserves intermediate robust features while reinitializing incompatible layers. Our experimental results demonstrate that robust initialization consistently improves performance under low-to-moderate label noise, accelerates early-stage learning, and degrades more gracefully as noise increases compared to random initialization. These findings provide empirical evidence that robustness learned against spurious correlations can generalize across domains and act as an effective inductive bias for learning with noisy labels.

INDEX TERMS Noisy Labels, Spurious Correlations, Distributionally Robust Optimization (DRO), Deep Learning, Computer Vision.

I. INTRODUCTION

Deep learning models deployed in the wild face a multitude of data quality challenges. The standard training paradigm, Empirical Risk Minimization (ERM), assumes that training data is clean and independently and identically distributed (i.i.d.) with the test data. However, real-world data is fraught with imperfections: group distributions are often imbalanced, leading to poor performance on underrepresented subgroups; labels are frequently noisy or incorrect, misleading the model during optimization; and domain shifts occur when the visual environment changes between training and deployment.

One particularly insidious issue is the problem of spurious correlations. Models trained via ERM often latch onto "shortcut" features such as the background texture, rather than the object of interest. For example, in the Waterbirds dataset, a model might predict "waterbird" simply because it detects water in the background. While this strategy maximizes average accuracy, it fails catastrophically when the correlation breaks (e.g., a waterbird on land). Similarly, noisy la-

bels—incorrect annotations in large-scale datasets—degrade performance by encouraging the model to memorize incorrect mappings, further distorting the learned feature representation.

Current research often treats these issues as distinct. Yu et al. [5] addressed label noise under domain shift but did not explicitly account for minority groups. Conversely, Guo et al. [4] tackled noisy labels in imbalanced datasets using conditional distributionally robust optimization (DRO), yet their approach remains sensitive to domain shifts. This leaves a critical gap: Can mechanisms designed to fix spurious correlations also help mitigate label noise?

We hypothesize that a model pre-trained to disregard false backgrounds (via Group DRO) will acquire invariant object properties. When this model is applied to a noisy label task, these invariant features will inhibit the model from overfitting to the noise, offering a more stable initialization for noise-correction techniques such as DivideMix [2].

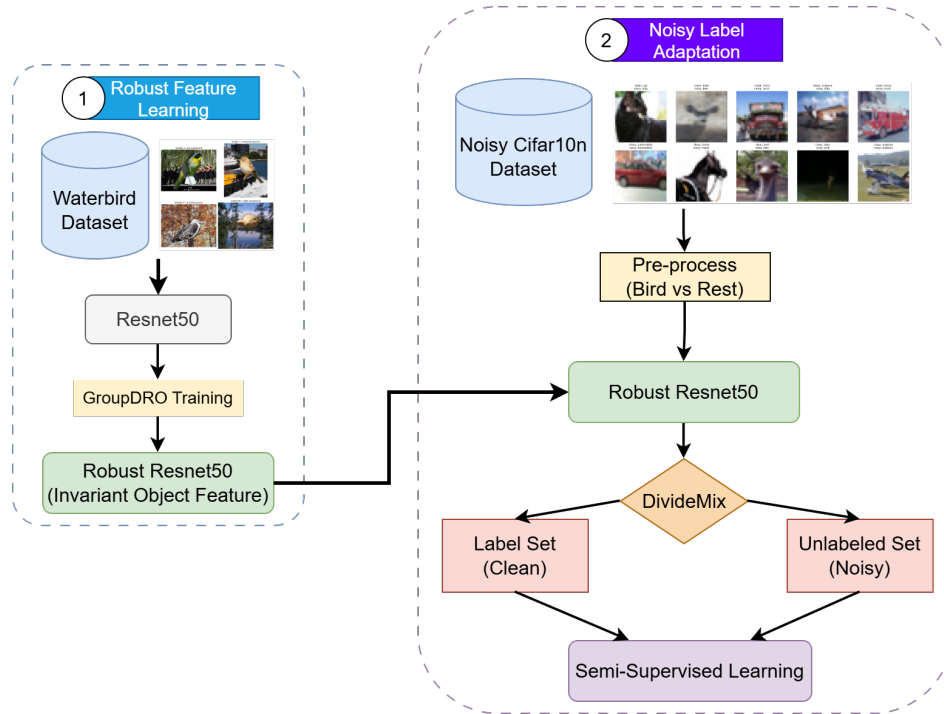


FIGURE 1: Overview of the proposed framework combining robust feature learning with noisy label adaptation.

II. LITERATURE REVIEW

Deep learning has seen many studies about its robustness within numerous different problems (noisy labels, group imbalances, etc.) but most have been focused upon one type of problem at a time; therefore, there is a great deal of motivation for research that will provide a unified framework to be able to solve multiple types of problems of this sort together.

A. LEARNING WITH NOISY LABELS

Noisy labels are a widespread phenomenon in real world datasets as a result of human error in the data's annotations, ambiguity in the labels or using an automated method to assign labels. The ultimate behavior of deep neural networks when faced with noisy labels is to "learn" the noisy labels as well and therefore degrade the performance of the network on unseen examples [8]. In a paper that provides a complete review of all types of methods for learning with noisy labels, [8] provide a classification of methods for learning with noisy labels into four categories: robust loss function, sample selection strategy, re-weighted samples and meta-learners. The goal of using a robust loss function is to minimize the impact of incorrect labels on the model's performance. Sample selection strategies take advantage of the fact that models tend to initially learn clean examples before they learn noisy ones.

An example of a sample selection strategy would be Co-teaching, which was first introduced by Han et al. [9], in which two networks are co-trained and share small-loss examples during training. This sharing of samples allows each network to filter out the erroneous labels of the other network, and thus is a very effective way to minimize the negative

effects of noisy labels on model performance. Although Co-teaching has been shown to be highly effective in noisy CIFAR-type environments, it does not address the issues of either domain shift or group imbalance.

B. GROUP INEQUALITY AND DISTRIBUTIONALLY ROBUST OPTIMIZATION

Distributionally Robust Optimization (DRO) has become an important tool in reducing the disparity of predictive performance among data subsets. As opposed to ERM's goal of optimizing the average loss, DRO seeks to optimize for the worst case loss, constrained to certain predefined groups. The authors in Sagawa et al. demonstrated the potential of Group DRO in addressing spurious correlations, specifically on the Waterbirds dataset, where the background information is dominant when making a prediction decision [1].

Later extensions to Group DRO have been made in order to represent more realistic use cases. The authors in Sagawa et al. propose Probabilistic [1]. PG-DRO allows for the possibility of probabilistically assigning members into one of the predefined groups, relaxing the constraints placed on the group assignment from being known and fixed. While DRO-based approaches are successful in improving the performance of the minority group, they rely upon having clean labels and do not account for the presence of label noise.

C. INTERSECTIONS OF DRO AND NOISY-LABEL ROBUSTNESS

In recent years, research has been conducted to investigate relationships between noisy-label learning and distribution-

ally robust optimization. The authors in Zhu et al., propose Label-Distributionally Robust (LDR) losses [11]. LDR losses model the uncertainty associated with the label distributions via a DRO formulation. Their results demonstrate that the robustness of a model to label noise and class imbalance can be obtained through the design of loss functions based on label-distributional uncertainty.

The authors in Guo et al [4]. expand upon the authors' previous idea of modeling the learning process with noisy labels as a Conditional Distributionally Robust Optimization (CDRO) problem [4]. CDRO models the uncertainty associated with the conditional label distribution and demonstrates both theoretically and empirically improved results compared to other techniques under label noise [4]. Both works provide evidence that DRO is a promising framework for incorporating noisy labels; however, both works address label uncertainty rather than domain-shift.

III. METHODOLOGY

Figure 1 illustrates our proposed framework, which operates in two distinct stages. First, we employ Group DRO to learn robust features on a source dataset characterized by known spurious correlations (Waterbirds). Subsequently, we transfer this model to a target dataset containing noisy labels (Noisy CIFAR-10N) and fine-tune it using a noise-robust algorithm.

A. PHASE 1: ROBUST FEATURE PRE-TRAINING

The Waterbirds dataset serves as a standard benchmark for evaluating spurious correlations and worst-group accuracy.

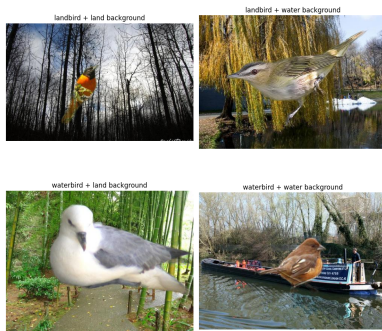


FIGURE 2: Examples from the Waterbirds dataset illustrating different background contexts.

Figure 2 presents examples from the dataset, highlighting the discrepancies between the target object (bird) and the background context.

Furthermore, the sample distribution across groups and backgrounds is significantly misbalanced, making this dataset ideal for evaluating Distributionally Robust Optimization (DRO) methods. As shown in Figure 3, the data distribution exhibits high class and group imbalance.

We utilized the Waterbirds dataset, which highlights disparities caused by spurious correlations between objects and backgrounds [1]. The training objective is to minimize the

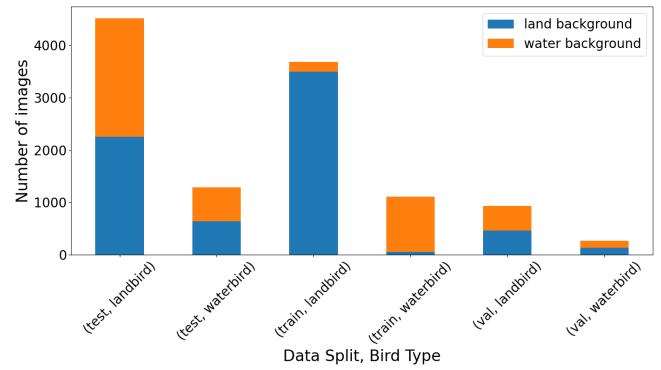


FIGURE 3: Distribution of samples per group in the Waterbirds dataset.

worst-group risk. Equation (1) defines the Group Distributionally Robust Optimization (GroupDRO) objective. Unlike standard Empirical Risk Minimization (ERM), which minimizes the average loss over the entire dataset, often allowing the model to ignore minority groups this objective focuses on the worst-case scenario,

$$\mathcal{L}_{DRO}(\theta) = \max_{g \in \mathcal{G}} \mathbb{E}_{(x,y) \sim P_g} [\ell(\theta; (x, y))] \quad (1)$$

For each distinct group g in the set $\mathcal{G}$, the term $\mathbb{E}_{(x,y) \sim P_g}$ calculates the expected loss purely within that group's distribution. The max operator then identifies the specific group for which the model currently has the highest loss. By minimizing this maximum value, the optimization process forces the model to improve its performance on the hardest group (e.g., waterbirds on land), thereby preventing it from relying on spurious background correlations that only hold for the majority groups.

B. PHASE 2: TRANSFER TO NOISY ENVIRONMENT

After Phase 1, we obtain a model trained on the spurious correlation structure of the Waterbirds dataset. In Phase 2, we transfer the resulting robust encoder to CIFAR-10N, a variant of CIFAR-10 augmented with controlled, human-annotated label noise. This phase evaluates the transferability of robust features from a spurious correlation setting (Waterbirds) to a noisy label setting.

The CIFAR-10N dataset contains over 50,000 training samples across 10 classes, with approximately 10% noisy labels relative to the base dataset. Figure 4 shows the number of noisy and clean labels present in the current dataset.

1) Target-Specific Binarization

To align the dataset with our specific target of bird detection, we binarized the multi-class labels. Rather than evaluating multi-class noisy-label learning, we deliberately reformulate CIFAR-10N into a highly imbalanced binary task to isolate whether robust representations can recover a rare semantic signal under label noise. All samples originally belonging to the "Bird" class were mapped to the positive class ($y = 1$),

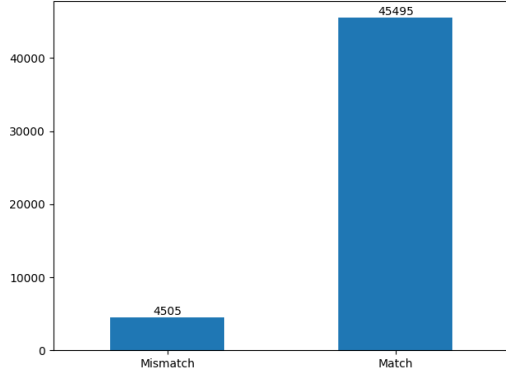


FIGURE 4: Distribution of noisy and clean labels in Noisy CIFAR-10N.

while samples from all other nine classes were mapped to the negative class ($y = 0$).

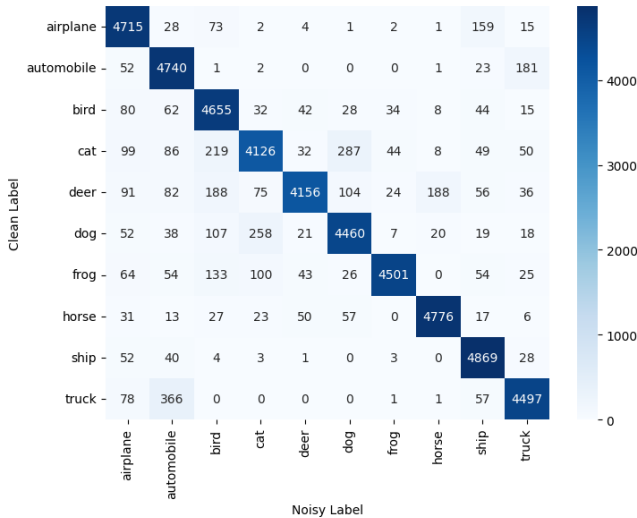


FIGURE 5: Clean and noise label distribution per class in Cifar-10 Dataset

This binarization process introduces significant class imbalance. As shown in Figure 5, the positive "Bird" class consists of only 4,655 samples, representing approximately 9.3% of the total dataset. Figure 6 shows the noise and ground truth class distribution for bird vs rest at 0.5 noise label scenario.

2) Semi-Supervised Noise Adaptation

We employ a semi-supervised pipeline similar to DivideMix [2]. The process relies on a "Warmup" phase where the model trains on the dataset for a few epochs to learn easy patterns.

Usually, standard random initialization of the model often leads to memorization of noise during this phase. We replace the random initialization with our DRO-pretrained weights. We hypothesize that the robust features will allow the model to classify "clean" images correctly much faster than it can memorize "noisy" labels, creating a wider divergence in the loss distributions of clean vs. noisy samples.



FIGURE 6: Impact of 50% Symmetric Noise on Class Distribution

C. FINE-TUNING THE TRANSFERRED MODEL

A significant technical challenge is the domain and architectural mismatch between source and target tasks. Waterbirds images are high-resolution (224×224), processed by a standard ResNet-50 with a 7×7 initial convolution (stem). In contrast, CIFAR-10 images are low-resolution (32×32), requiring a modified ResNet backbone with a 3×3 initial convolution to preserve spatial dimensions. To enable effective transfer, we adapted the model as follows:

- 1) **Layer Mapping:** We mapped the semantic layers between the architectures, ensuring compatibility between the pre-trained weights and the target architecture (e.g., aligning downsample blocks with shortcut connections).
- 2) **Selective Filtering (Stem and Head Replacement):** We discarded the first convolutional layer (conv1, 7×7) and the final fully connected layer (linear, 2 classes) due to input resolution and output dimension mismatches. These layers were reinitialized.
- 3) **Feature Preservation:** We loaded weights for all intermediate residual blocks (Layers 1-4), preserving the complex, robust feature extractors learned via Group DRO.

D. EXPERIMENTAL SETUP

In Phase 1, we employed a ResNet-50 architecture trained on the Waterbirds dataset. We utilized a batch size of 64 and a learning rate of 0.001. The training duration was extended to 300 epochs to ensure the model sufficiently captured the spurious correlations and converged on the worst-group distributions within the source domain.

In Phase 2, we benchmarked our approach on the CIFAR-10N dataset with symmetric label noise. We compared two initialization strategies:

- **Baseline:** A standard ResNet-50 with random initialization.
- **Robust Transfer (Ours):** The ResNet-50 initialized with the robust weights learned via Group DRO in Phase 1.

For this downstream task, both models were fine-tuned for 10 epochs using a learning rate of 0.02 and a batch size of 128.

For both phases the Stochastic Gradient Descent (SGD) were used as an optimizer.

TABLE 1: Hyperparameter settings for Phase 1 (Pre-training) and Phase 2 (Transfer).

Hyperparameter	Phase 1 (Waterbirds)	Phase 2 (CIFAR-10N)
Architecture	ResNet-50	ResNet-50 (Modified)
Initialization	Pre-trained (ImageNet)	Random vs. Robust (DRO)
Epochs	300	10
Batch Size	64	128
Learning Rate	0.001	0.02
Optimizer	SGD	SGD

IV. RESULTS

This section presents the experimental findings from both phases of our research: the initial robust feature learning on the Waterbirds dataset and the subsequent evaluation on the noisy CIFAR-10N benchmark. We further provide a comparative analysis of the proposed robust transfer method against the baseline across varying noise ratios.

A. ROBUST FEATURE LEARNING (GROUP DRO)

Following the methodology defined by Sagawa et al. [1], we employed Group Distributionally Robust Optimization (Group DRO) to train a ResNet-50 model on the Waterbirds dataset to mitigate reliance on spurious correlations. The training progression is illustrated in Figure 7.

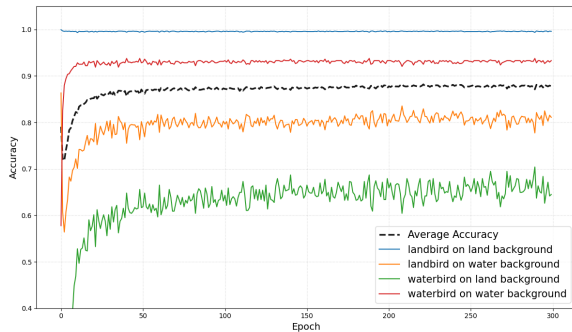


FIGURE 7: Group-wise accuracy over 300 epochs during Waterbirds pre-training.

As observed in figure 7, while the average accuracy is high from the outset, the worst-group accuracy starts poorly, indicating initial reliance on spurious background features. However, as training progresses, the model successfully optimizes for the worst-case groups. The final group-wise accuracies are detailed in Table 2, demonstrating that the model achieves respectable performance even on conflicting groups (e.g., Waterbirds on Land), thereby learning invariant object features.

TABLE 2: Final Group-wise Accuracy on Waterbirds (Epoch 300)

Group	Accuracy
Landbird on land background	99.6%
Landbird on water background	81.1%
Waterbird on land background	64.5%
Waterbird on water background	93.3%

B. QUANTITATIVE PERFORMANCE ON CIFAR-10N

We evaluated the transferability of these robust features to the CIFAR-10N dataset. Table 3 presents the final test accuracy for the standard noise setting ($r = 0.5$). The proposed robust initialization significantly outperforms the random initialization, validating the hypothesis that feature quality impacts noise resilience.

TABLE 3: Test Accuracy on CIFAR-10N ($r = 0.5$)

Method	Model Initialization	Test Acc (%)
Standard DivideMix (Baseline)	Random	90.20
Robust Transfer (Ours)	Waterbirds DRO	93.84

C. IMPACT OF VARYING NOISE RATIOS

To assess stability, we evaluated both the baseline (ResNet50) and our proposed model across a spectrum of noise ratios: 0.1, 0.3, 0.5, 0.7, 0.9.

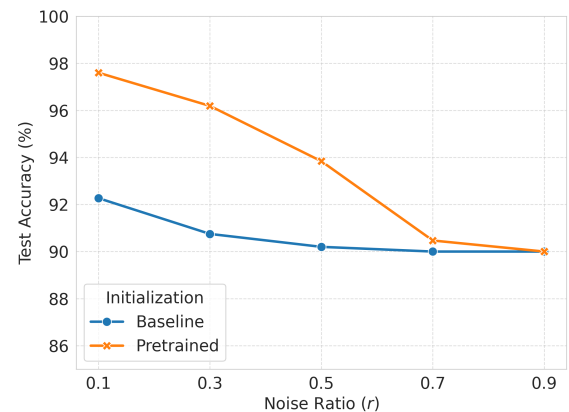


FIGURE 8: Impact of Noise Ratio on Test Accuracy.

Figure 8 illustrates the accuracy trends. The baseline model exhibits a stagnant performance hovering around 90% across almost all noise levels. In contrast, our pre-trained model maintains high accuracy at lower noise ratios (97.6% at $r = 0.1$) and degrades gracefully as noise increases. Notably, at the extreme noise ratio of 0.9, both models converge to the baseline performance, indicating that excessive label corruption overwhelms the benefits of robust initialization.

The advantage of our approach is further quantified in Figure 9, which plots the optimality gap, the performance gain achieved by using robust features over the baseline.

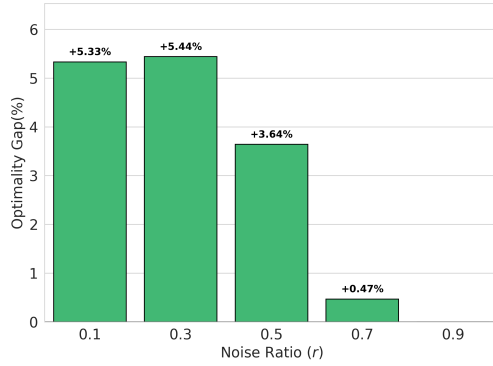
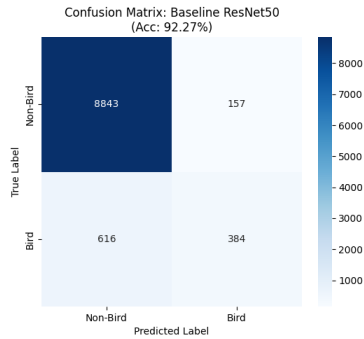


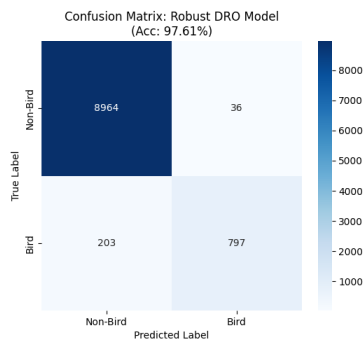
FIGURE 9: Optimality Gap: Performance Gain of Robust Features over Baseline.

D. CONFUSION MATRIX ANALYSIS

To investigate the source of the performance gap, we analyzed the confusion matrices at noise ratios of 0.1 (Figure 10) and 0.5 (Figure 11).



(a) Baseline (Random Init)

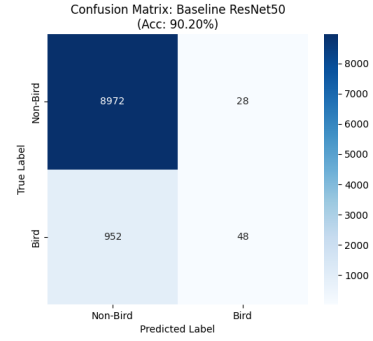


(b) Robust Transfer (Ours)

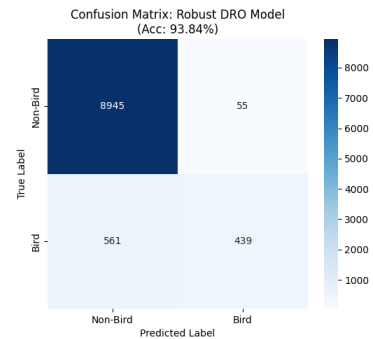
FIGURE 10: Comparison of confusion matrices at noise ratio $r = 0.1$.

V. DISCUSSION

The results highlight a critical failure mode in standard training under class imbalance and label noise. As observed in Figure 8, the baseline model's accuracy remains fixed at approximately 90%. A plausible explanation lies in the severe



(a) Baseline (Random Init)



(b) Robust Transfer (Ours)

FIGURE 11: Comparison of confusion matrices at noise ratio $r = 0.5$.

class imbalance induced by binarization. Since the target class ("Bird") constitutes only $\approx 10\%$ of the dataset, the baseline model fails to learn the minority signal amidst the noise and minimizes loss by simply predicting the majority class ("Non-Bird") for all inputs.

In contrast, our proposed model leverages the invariant features learned via Group DRO to escape this trap. By analyzing the confusion matrices in Figure 10 and Figure 11, we observe that the Robust Transfer model achieves a significantly higher rate of True Positives (correctly identified birds) compared to the baseline. While high noise eventually destabilizes the model ($r = 0.7, 0.9$), the superior performance at low-to-moderate noise levels confirms that robustness is a transferable property. By forcing the model to unlearn spurious background correlations in the source domain (Waterbirds), we effectively regularized the feature extractor. When applied to the target domain (CIFAR-10N), this regularization prevented the model from overfitting to the "noise" in the labels, much like it prevented overfitting to the "noise" in the backgrounds during pre-training.

VI. CONTRIBUTIONS

This work makes the following key contributions:

- **Robust Feature Transfer for Noisy Labels:** We demonstrate that features learned via Group Distributionally Robust Optimization (Group DRO) on a

spurious-correlation benchmark can improve robustness to label noise in a downstream task, establishing a novel connection between spurious-correlation mitigation and noisy-label learning.

- **Sequential Robust Transfer Framework:** We propose a two-stage training framework that first enforces group robustness in a source domain and then transfers the resulting invariant representations to a noisy-label target domain, bridging two previously isolated research directions.
- **Empirical Evidence of Transferable Robustness:** Through extensive experiments on Waterbirds and CIFAR-10N across varying noise ratios, we show that robust initialization improves performance under low-to-moderate label noise and degrades more gracefully than random initialization.
- **Practical Cross-Domain Adaptation Strategy:** We introduce a selective layer transfer mechanism that enables effective weight reuse across architectures with differing input resolutions, facilitating robust cross-domain transfer in practice.

VII. CONCLUSION AND FUTURE WORK

In this work, we investigated whether robustness learned through Distributionally Robust Optimization (DRO) against spurious correlations can transfer to learning scenarios dominated by noisy labels. By pretraining a ResNet-50 model using Group DRO on the Waterbirds dataset and transferring the resulting representation to CIFAR-10N, we demonstrated that robust feature initialization consistently improves performance under label noise. Our results show that robustness, when learned at the representation level, can act as an implicit regularizer that mitigates overfitting to corrupted labels in downstream tasks. Empirically, the proposed robust transfer approach outperforms standard random initialization across a wide range of noise ratios, particularly in low-to-moderate noise regimes where clean signal remains informative. These findings suggest that robustness is not task-specific but can be learned and reused across domains with fundamentally different failure modes.

FUTURE WORK

Several directions remain open for future research. First, we plan to extend this framework to additional backbone architectures (e.g., ResNet-18, Vision Transformers) to evaluate the generality of robust feature transfer. Second, a more exhaustive exploration of optimization parameters, including learning rates, batch sizes, optimizers, and warmup schedules, could further elucidate the interaction between robust initialization and noisy-label learning dynamics. Finally, future work will investigate cross-domain robustness more explicitly by transferring from multiple spurious-correlation datasets to diverse noisy-label benchmarks, as well as exploring joint training objectives that simultaneously model spurious correlations and label uncertainty.

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liability and security of modern power grids.

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